4. Use word embeddings to improve prompts for Generative AI model. Retrieve similar words using word embeddings. Use the similar words to enrich a GenAI prompt. Use the AI model to generate responses for the original and enriched prompts. Compare the outputs in terms of detail and relevance.

```
pip install gensim

pip install nltk
pip install transformers
```

```
from gensim.downloader import load
from transformers import pipeline
import nltk
import string
from nltk.tokenize import word_tokenize
nltk.download('punkt_tab')
print("loading pre trained word vectors")
word_vectors=load("glove-wiki-gigaword-100")

def replace_keyword_in_prompt(prompt,keyword,word_vectors,topn=1):
  words=word_tokenize(prompt)
  enriched_words=[]
  for word in words:
    cleaned_word=word.lower().strip(string.punctuation)
    if cleaned_word==keyword.lower():
      try:
        similar_words=word_vectors.most_similar(cleaned_word,topn=topn)
        if similar_words:
          replacement_word=similar_words[0][0]
          print(f"Replacing {word}-> {replacement_word}")
          enriched_words.append(replacement_word)
          continue
      except KeyError:
        print(f"{keyword} not found in vocabulary using original word")
        enriched_words.append(word)
  enriched_prompt=" ".join(enriched_words)
  print(f"\n Enriched Prompt:{enriched_prompt}")
  return enriched_prompt
print("\n Loading GPT-4 model")
```

```
generator=pipeline("text-generation",model="gpt2")

def generate_response(prompt,max_length=100):
  try:

        response=generator(prompt,max_length=max_length,num_return_sequen
        ces=1)
        return response[0]['generated_text']
  except Exception as e:
    print(f"error generating response {e}")
    return None
original_prompt="write an essay on natural disaster"
print(f"Original prompt: {original_prompt}")
k_term="disaster"
enriched_prompt=replace_keyword_in_prompt(original_prompt,k_term,word_v
ectors)
print("\n generating response for original prompt")
original_response=generate_response(original_prompt)
print(original_response)

print("\n generating response for enriched prompt")
enriched_response=generate_response(enriched_prompt)
print(enriched_response)

print("\n comparison of responses")
print("original prompt response length",len(original_response))
print("enriched prompt response length",len(enriched_response))
print("original prompt response detail",original_response.count("."))
print("enriched prompt response detail",enriched_response.count("."))
```